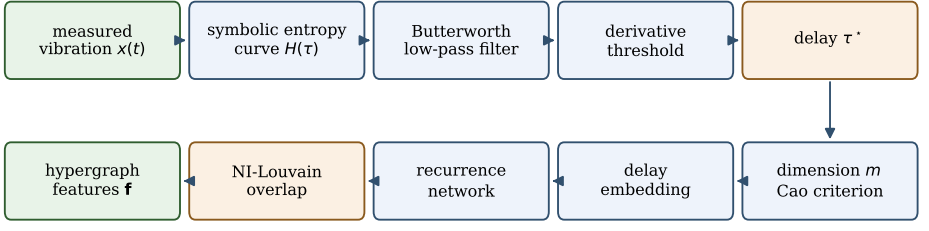
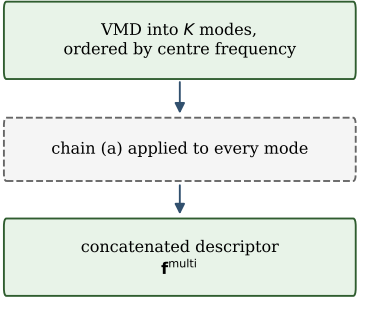
(a) single-resolution chain



(b) multi-resolution extension